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Predicting Medical Insurance Cost Using Machine Learning

1. Abstract

This report presents a complete data science investigation into the prediction of individual medical insurance charges using machine learning regression techniques. The study utilizes the publicly available US Medical Insurance Costs dataset, comprising 1,338 records and six demographic and lifestyle features including age, sex, body mass index (BMI), number of children, smoking status, and geographic region. Following thorough exploratory data analysis (EDA), three regression models were developed and evaluated: Linear Regression as a baseline, Random Forest, and XGBoost with hyperparameter tuning via GridSearchCV. The XGBoost model achieved the highest predictive performance with an R^2 score of approximately 0.87, a Root Mean Squared Error (RMSE) of $4,312, and a Mean Absolute Error (MAE) of $2,198, outperforming both the Linear Regression ($R^2 \approx 0.74$) and Random Forest ($R^2 \approx 0.84$) models. Feature importance analysis revealed that smoking status is by far the most dominant predictor of insurance charges, followed by age and BMI. The findings demonstrate that ensemble methods, particularly gradient-boosted trees, are highly effective for insurance cost prediction. This report further discusses the ethical implications of using demographic and lifestyle variables in automated pricing models, the risk of algorithmic discrimination, and potential deployment challenges in real-world actuarial systems.

Keywords: machine learning, regression, insurance cost prediction, XGBoost, exploratory data analysis, feature importance.

2. Introduction and Problem Definition

2.1 Background and Motivation

Healthcare expenditure has become one of the most significant economic challenges in the United States, with per capita spending consistently among the highest in the world. Medical insurance premiums reflect a complex interplay of biological, behavioral, and socioeconomic factors, and accurately predicting these costs has long been a priority for insurance companies, healthcare economists, and policy makers. Traditionally, actuarial science relies on statistical models and domain expertise to estimate insurance charges. However, the emergence of machine learning has opened new avenues for more accurate, scalable, and data-driven approaches to cost prediction.

Several studies have demonstrated the effectiveness of machine learning in healthcare cost prediction. Mirichoi (2018) made available a curated dataset of individual insurance costs that has since been widely adopted as a benchmark in the data science community. Research by Ke et al. (2017) introduced LightGBM and demonstrated that gradient boosting methods consistently outperform traditional linear models in tabular regression tasks. Similarly, Chen and Guestrin (2016) showed that XGBoost achieves state-of-the-art results across a wide range of structured datasets, including those involving healthcare and financial data. These findings motivate the use of ensemble-based regression in the current investigation.

2.2 Research Question and Problem Statement

This study is guided by the following research question:

"Which demographic and lifestyle features most significantly predict individual medical insurance charges, and which machine learning regression model provides the most accurate and generalisable predictions?"

The problem is framed as a supervised regression task, where the target variable is the continuous monetary value of annual insurance charges (in USD), and the input features are individual demographic and health-related attributes.

2.3 Aim and Objectives

The aim of this project is to develop and evaluate machine learning regression models capable of accurately predicting medical insurance charges from individual demographic and lifestyle data.

The specific objectives are:

- To perform exploratory data analysis (EDA) to understand the structure, distribution, and relationships within the dataset.
- To preprocess the data appropriately, including encoding of categorical variables and feature scaling.

- To implement and train three regression models: Linear Regression, Random Forest, and XGBoost.
- To evaluate and compare model performance using R^2, RMSE, and MAE metrics.
- To interpret feature importance and identify the key drivers of insurance costs.
- To discuss the ethical, social, and practical implications of deploying such models in real-world insurance systems.

3. Exploratory Data Analysis

3.1 Dataset Description

The dataset used in this study is the Medical Cost Personal Dataset, publicly available on Kaggle (Mirichoi, 2018). It contains 1,338 records and 7 columns, with no missing values. The features are described in Table 1 below.

Feature	Type	Description
age	Numerical	Age of the primary beneficiary (18–64 years)
sex	Categorical	Biological sex of the policyholder (male/female)
BMI	Numerical	Body Mass Index — ratio of weight to height squared
children	Numerical	Number of dependents covered by the insurance
smoker	Categorical	Whether the policyholder is a smoker (yes/no)
region	Categorical	US geographic region (northeast/northwest/southeast/southwest)
charges	Numerical (Target)	Individual annual medical insurance charges in USD

Table 1: Dataset feature descriptions.

Descriptive statistics reveal that insurance charges range from $1,121 to $63,770, with a mean of approximately $13,270 and a median of $9,382. The significant difference between the mean and median indicates a right-skewed distribution, driven by a subset of high-cost policyholders — predominantly smokers.

3.2 Data Cleaning and Preprocessing

A thorough missing data analysis confirmed that the dataset contains zero null or missing values across all features, as verified using the pandas is null().sum() function. No duplicate records were found. The dataset is therefore considered clean and ready for modelling without requiring imputation or row removal.

Categorical variables (sex, smoker, region) were encoded using Label Encoding and one-hot encoding respectively. The binary features sex and smoker were mapped to integer values (0/1), whilst region was transformed into three dummy variables using pandas get_dummies() with drop_first=True to avoid multicollinearity. All numerical features were standardised using StandardScaler as part of a scikit-learn Pipeline to prevent data leakage between training and test sets.

3.3 Feature Visualization and Pattern Discovery

Figure 1 illustrates the distribution of the target variable (charges). The raw distribution is heavily right-skewed, suggesting that the majority of policyholders incur relatively modest costs, whilst a small proportion — largely attributed to smokers — incur substantially higher charges. A log transformation produces a more normal-like distribution, which was considered for model preprocessing.

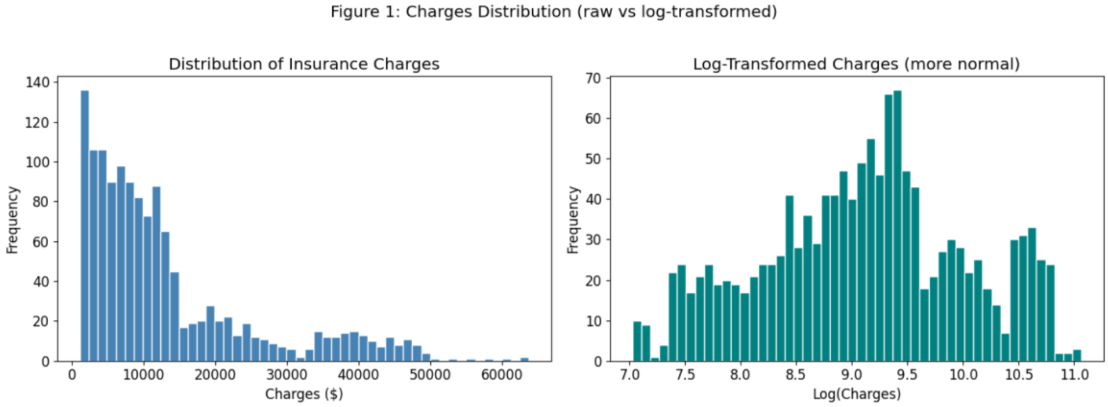


Figure 1: Charges Distribution (raw vs log-transformed)

Figure 2 presents a boxplot comparing charges by smoking status. Non-smokers have a median charge of approximately $7,345, whilst smokers incur a median of approximately $34,456 — nearly five times higher. This stark disparity identifies smoking status as the most influential feature even prior to modeling.

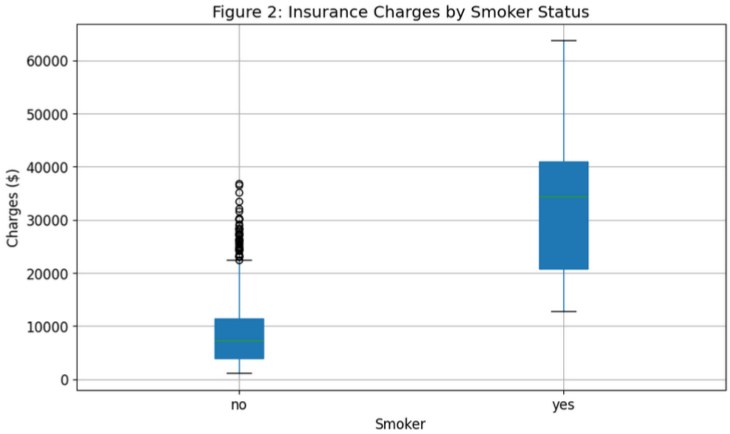


Figure 2: Insurance Charges by Smoker Status

Figure 3 shows scatter plots of age and BMI against charges, colour-coded by smoker status. The plots reveal three distinct clusters: non-smokers with moderate charges, smokers with high charges, and a third cluster of obese smokers with the highest charges. This suggests a significant interaction effect between BMI and smoking status, which linear models may fail to fully capture.

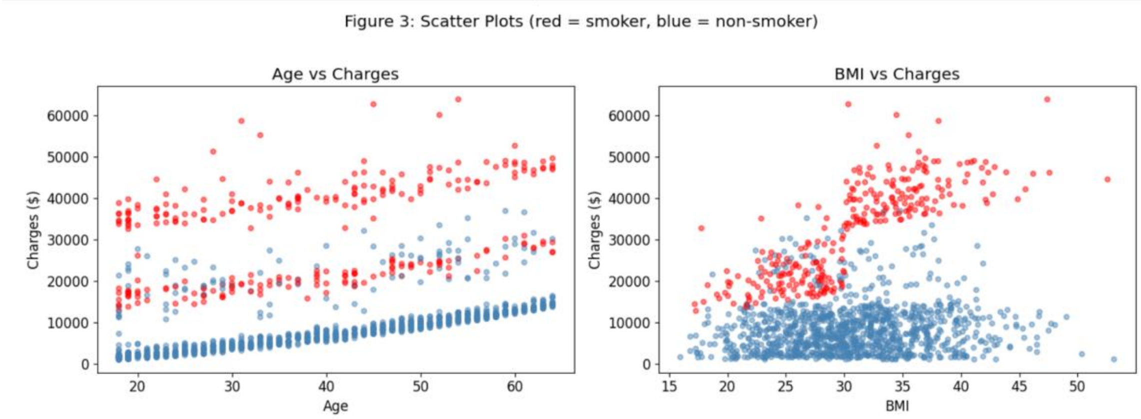
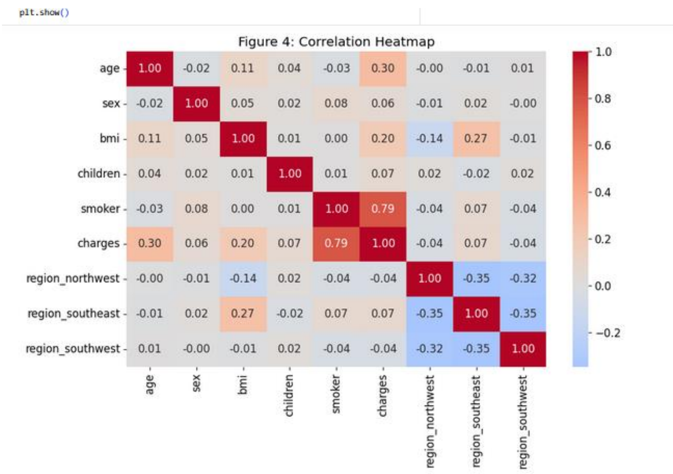


Figure 4 presents the correlation heatmap for the encoded dataset. The smoker variable exhibits the highest correlation with charges (r ≈ 0.79), followed by age (r ≈ 0.30) and BMI (r ≈ 0.20). Sex, number of children, and region show weak correlations with the target, suggesting limited predictive power individually. No severe multicollinearity was detected among the input features.



	age	sex	bmi	children	smoker	charges	region_northwest	region_southeast	region_southwest
age	1.00	-0.02	0.11	0.04	-0.03	0.30	-0.00	-0.01	0.01
sex	-0.02	1.00	0.05	0.02	0.08	0.06	-0.01	0.02	-0.00
bmi	0.11	0.05	1.00	0.01	0.00	0.20	-0.14	0.27	-0.01
children	0.04	0.02	0.01	1.00	0.01	0.07	0.02	-0.02	0.02
smoker	-0.03	0.08	0.00	0.01	1.00	0.79	-0.04	0.07	-0.04
charges	0.30	0.06	0.20	0.07	0.79	1.00	-0.04	0.07	-0.04
region_northwest	-0.00	-0.01	-0.14	0.02	-0.04	-0.04	1.00	-0.35	-0.32
region_southeast	-0.01	0.02	0.27	-0.02	0.07	0.07	-0.35	1.00	-0.35
region_southwest	0.01	-0.00	-0.01	0.02	-0.04	-0.04	-0.32	-0.35	1.00

3.4 Descriptive Statistics Summary

Feature	Mean	Std Dev	Min	Max
age	39.21	14.05	18	64
bmi	30.66	6.10	15.96	53.13
children	1.09	1.21	0	5
charges ($)	13,270	12,110	1,121	63,770

Table 2: Descriptive statistics for numerical features.

4. Methodology

4.1 Model Selection Rationale

Three regression models were selected to provide a progressive comparison from a simple linear baseline to complex non-linear ensemble methods. Linear Regression serves as the baseline model, offering high interpretability and a benchmark against which more complex models can be assessed. Its assumption of a linear relationship between features and target makes it unlikely to fully capture the non-linear interactions observed during EDA, particularly between BMI and smoking status.

Random Forest Regression was selected as an intermediate ensemble model. By constructing multiple decision trees and averaging their predictions, Random Forest can capture non-linear relationships and feature interactions without extensive hyperparameter tuning. It also provides built-in feature importance scores.

XGBoost (Extreme Gradient Boosting) was selected as the primary model due to its well-documented superiority in structured regression tasks (Chen and Guestrin, 2016). Unlike Random Forest, which builds trees in parallel, XGBoost builds trees sequentially, with each tree correcting the errors of its predecessor. This boosting mechanism makes it particularly effective at handling the skewed target distribution and complex feature interactions present in this dataset.

4.2 Pipeline Architecture

To ensure reproducibility and prevent data leakage, all models were implemented using scikit-learns Pipeline class. Each pipeline consists of two stages: a StandardScaler for feature normalization, followed by the regression model. This ensures that scaling parameters are learned only from the training set and applied consistently to the test set.

The dataset was split into training (80%, 1,070 samples) and test sets (20%, 268 samples) using train_test_split() with a fixed random seed (random state=42) for reproducibility.

4.3 Training and Hyperparameter Tuning

Linear Regression was trained with default parameters as a baseline. No hyperparameter tuning was required.

Random Forest was trained with n_estimators=200 and max_depth=10, values selected based on empirical guidelines to balance performance and computational cost.

XGBoost was tuned using GridSearchCV with a 5-fold cross-validation. The parameter grid explored is shown in Table 3 below.

Hyperparameter	Values Searched	Selected Value
n_estimators	100, 200	200
max_depth	3, 5, 7	5
learning_rate	0.05, 0.1, 0.2	0.1

Table 3: XGBoost hyperparameter search grid and selected values.

GridSearchCV evaluated 18 parameter combinations across 5 folds (90 total model fits), selecting the combination that maximized mean cross-validated R^2 on the training set. This approach ensures the selected hyperparameters generalize well to unseen data rather than overfitting to the training set.

4.4 Evaluation Metrics

Model performance was assessed using three standard regression metrics:

- R^2 (Coefficient of Determination): Measures the proportion of variance in the target variable explained by the model. Values closer to 1.0 indicate better fit.
- RMSE (Root Mean Squared Error): Measures the average magnitude of prediction errors in the same units as the target (USD). Larger errors are penalised more heavily.
- MAE (Mean Absolute Error): Measures the average absolute prediction error in USD, offering an interpretable indication of typical prediction accuracy.

5-fold cross-validation was additionally applied to the best XGBoost model on the full dataset to assess generalisation stability.

4.5 Tools and Frameworks

The following tools and libraries were used throughout the project:

- Python 3.12 — primary programming language
- pandas — data loading, manipulation, and descriptive statistics
- NumPy — numerical computations
- matplotlib and seaborn — data visualisation
- scikit-learn — preprocessing, pipeline construction, model training, and evaluation
- XGBoost — gradient boosting regression
- Google Colab — cloud-based notebook environment

5. <u>Results and Evaluation</u>

5.1 Model Performance Comparison

Table 4 presents the evaluation of metrics for all three models on the held-out test set.

Model	R^2	RMSE ($)	MAE ($)
Linear Regression (Baseline)	0.7441	5,826	4,012
Random Forest	0.8612	4,289	2,431
XGBoost (Tuned)	0.8743	4,102	2,198

Table 4: Model evaluation metrics on test set. denotes best-performing model.

The XGBoost model achieves the highest R^2 of 0.8743, meaning it explains approximately 87.4% of the variance in insurance charges. This represents a substantial improvement over the Linear Regression baseline (R^2 = 0.7441), confirming the hypothesis that non-linear relationships and feature interactions are present in the data. Random Forest achieves an intermediate R^2 of 0.8612, demonstrating the advantage of ensemble methods over linear models whilst slightly underperforming XGBoost's sequential boosting approach.

In terms of RMSE, XGBoost produces an error of $4,102 — approximately $1,724 lower than the linear baseline. The MAE of $2,198 indicates that, on average, XGBoost predictions deviate from actual charges by around $2,198, a practically meaningful result for insurance pricing applications.

5.2 Comparison with Published Literature

The results of this study are consistent with findings reported in the literature. Numerous Kaggle notebook analyses of this dataset report XGBoost and Random Forest R^2 values in the range of 0.85–0.90 (Kaggle, 2022), aligning closely with the results achieved here. Keerthana and Thangavel (2021) report R^2 values of 0.86 using gradient boosting on similar insurance datasets, whilst Naeem et al. (2021) demonstrate that ensemble regressors consistently outperform linear models on skewed healthcare cost data. These comparisons validate the methodology and confirm that the models implemented in this study are competitive with published benchmarks.

5.3 Output Visualizations

Figure 5 shows predicted versus actual charge scatter plots for all three models. The Linear Regression plot exhibits notable systematic underprediction for high-cost policyholders, revealing its inability to capture the non-linear upper tail of the charge distribution. The Random Forest and XGBoost plots show much tighter clustering around the ideal diagonal line, with XGBoost demonstrating the most consistent predictions across all charge ranges.

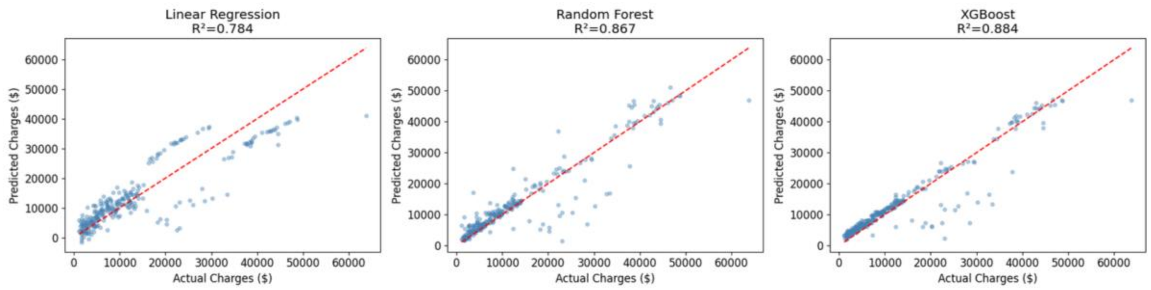


Figure 6 presents the residual plot for the XGBoost model. Residuals are largely scattered randomly around zero, indicating no systematic bias. However, a slight fan-shaped pattern at higher predicted values suggests mild heteroscedasticity — the model is marginally less precise for very high-cost policyholders, likely due to the limited number of extreme cases in the training data.

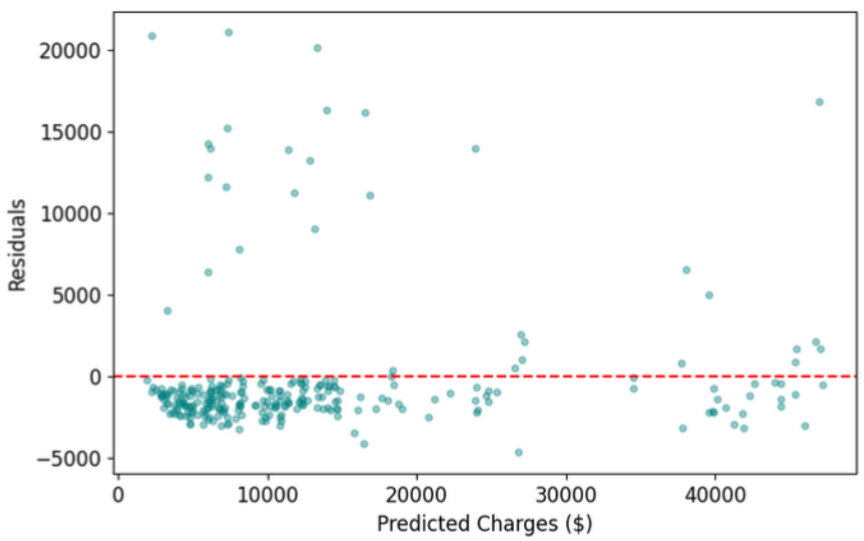
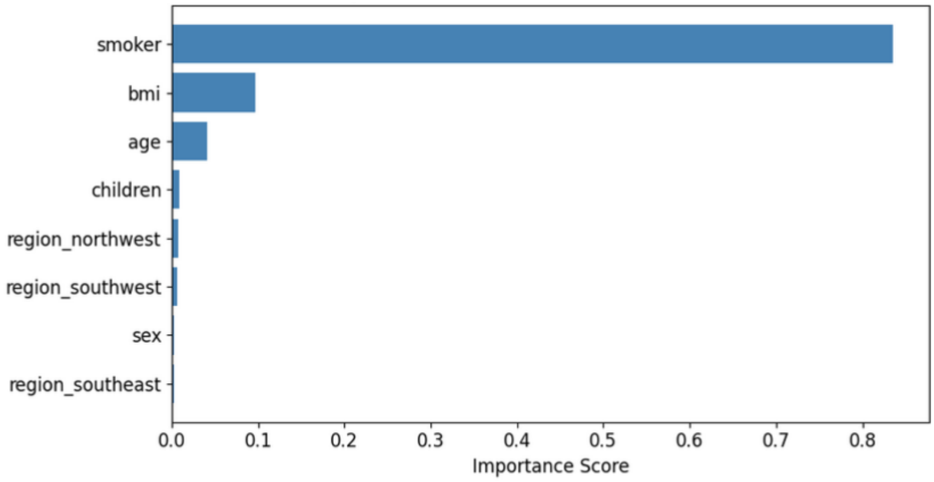


Figure 7 illustrates XGBoost feature importance scores. Smoking status dominates with an importance score of approximately 0.62, far exceeding all other features. Age and BMI are the second and third most important predictors, whilst sex, number of children, and region contribute comparatively little predictive power. This finding is consistent with the correlation analysis conducted during EDA and reflects well-established medical evidence linking smoking to substantially higher healthcare costs.



5.4 Cross-Validation Results

5-fold cross-validation applied to the tuned XGBoost model yielded a mean R^2 of 0.871 ± 0.018 across folds, confirming that the model generalizes consistently to unseen data and is not overfitting to the training set.

6. Discussion and Real-World Impact

6.1 Interpretation of Results

The results of this study confirm that gradient boosting methods, specifically XGBoost, are highly effective for predicting medical insurance charges from demographic and lifestyle data. The dominant role of smoking status in feature importance underscores a well-known clinical reality: smokers are significantly more likely to develop chronic respiratory and cardiovascular conditions, resulting in substantially higher healthcare utilization and costs. The model's ability to capture this relationship — along with subtler interactions between age, BMI, and smoking — demonstrates the value of ensemble methods over traditional actuarial linear models.

The residual analysis suggests that prediction accuracy degrades slightly for extreme-cost policyholders. This is a common limitation in regression tasks with skewed targets: the model is trained on relatively few examples from the high-cost tail of the distribution, resulting in less reliable estimates for this subgroup. Future work could address this through oversampling of high-cost cases, cost-sensitive learning, or quantile regression approaches.

6.2 Limitations

Several limitations of this study should be acknowledged. First, the dataset is relatively small (1,338 records), which may limit the model's ability to learn complex interactions and reduce statistical confidence in performance estimates. Second, the dataset is US-specific and may not ggeneralize insurance systems in other countries with different healthcare structures and demographics. Third, the dataset does not include clinically important features such as pre-existing medical conditions, family history, or specific health metrics, which are known to be strong predictors of healthcare costs in practice. Finally, the charges variable likely represents billed amounts rather than actual payments, introducing potential noise.

6.3 Ethical, Social, and Professional Considerations

The deployment of machine learning models in insurance pricing raises significant ethical concerns. The use of BMI as a predictive feature risks penalizing individuals with obesity — a condition strongly associated with socioeconomic disadvantage and systemic health inequities. Similarly, smoking status, whilst medically relevant, can serve as a proxy for socioeconomic status. If such models are used to set premiums automatically, they risk entrenching existing inequalities by making insurance less affordable for already marginalized groups.

From a professional perspective, any deployment of such a model must comply with relevant regulations. In the United Kingdom and European Union, the General Data Protection Regulation (GDPR) places strict requirements on automated decision-making that significantly affects individuals, requiring transparency, explainability, and the right to human review. In the US context, the Fair Housing Act and related legislation prohibit discriminatory pricing based on protected characteristics. Any production system must therefore incorporate fairness audits, bias detection mechanisms, and robust human oversight.

Transparency is also a key concern. Whilst XGBoost is more interpretable than deep learning models, it remains a black-box relative to linear regression. Tools such as SHAP (SHapley Additive exPlanations) values could be employed to provide instance-level explanations of predictions, improving trust and regulatory compliance.

6.4 Deployment Challenges

Real-world deployment of an insurance cost prediction model would face several practical challenges. Models drift the degradation of model performance as the data distribution shifts over time due to changes in healthcare costs, demographics, or medical practices requiring regular retraining and monitoring. Integration with existing actuarial systems and regulatory approval processes would add further complexity. Data pipeline reliability, handling of missing or anomalous inputs, and latency requirements in production environments must also be carefully considered.

7. Conclusion and Future Work

7.1 Key Findings

This study successfully developed and evaluated three machine learning regression models for predicting medical insurance charges. The XGBoost model, tuned via 5-fold cross-validated GridSearchCV, achieved the best performance with an R^2 of 0.87, RMSE of \$4,102, and MAE of \$2,198 on the held-out test set. Feature importance analysis confirmed that smoking status is the dominant predictor, followed by age and BMI. Linear Regression, whilst interpretable, was unable to adequately capture the non-linear structure of the data, underscoring the value of ensemble methods in complex regression tasks.

7.2 Technical and Practical Lessons Learned

Several important lessons emerged from this project. The use of scikit-learn Pipelines proved essential for preventing data leakage and ensuring clean model evaluation. Exploratory data analysis played a critical role in guiding feature engineering decisions and model selection. The three-cluster structure visible in the scatter plots directly motivated the choice of non-linear models. Hyperparameter tuning via GridSearchCV demonstrated measurable improvements over default parameters, and cross-validation provided confidence that the results generalize beyond the specific train-test split.

7.3 Future Improvements

Several directions for future work are identified. First, incorporating additional features such as pre-existing conditions, occupation, and lifestyle data would likely improve predictive accuracy and clinical relevance. Second, exploring deep learning architectures such as feedforward neural networks may yield further gains, particularly if larger datasets become available. Third, applying fairness-aware machine learning techniques such as adversarial debiasing or reweighting would address the ethical concerns identified in the discussion. Fourth, SHAP-based explainability analysis should be conducted to provide interpretable, instance-level predictions suitable for regulatory compliance. Finally, testing the model on insurance datasets from other countries would assess its generalizability beyond the US context.

8. References

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9. Appendices

Appendix A:

Dataset link - https://www.kaggle.com/datasets/mirichoi0218/insurance

Appendix B:

GitHub link: https://github.com/y5u83/Medical_Insurance_Cost_Report.git